DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology
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Research Interests

I study **motion** as a computational medium in computer graphics and human–computer interaction (HCI). I build interactive systems for designing, generating, and fabricating motion, connecting expressive intent with computational design and physical realization. My research interests include computer animation, computational design, motion synthesis, and AI-assisted creative tools. My earlier work spans machine-learning systems, cloud software, games, and social robots.

Education

Georgia Tech, Ph.D. in Computer Science — Aug. 2022–Present
- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE — Mar. 2020–Feb. 2022
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE — Mar. 2012–Feb. 2020

Experience

Ph.D. Research Intern, Dolby Laboratories — May 2024–Aug. 2025
- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. — June–Aug. 2023
- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. — Jan. 2021–Aug. 2022
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador — 2020–2022
- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc. — Jan.–Dec. 2019
- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio) — Apr. 2017–Jan. 2019
- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

1 **DoangJoo Synn**, Zhifan Guo, Sehoon Ha, and HyunJoo Oh. "MotionSmith: A Sketch-Based Design System for Automata Making." *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi:10.1145/3772318.3791545.

2 Jin Yu, Poojita Garg, **DoangJoo Synn**, and HyunJoo Oh. "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning." *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi:10.1145/3706598.3713260.

3 **DoangJoo Synn**, XinYu Piao, JooYoung Park, and Jong-Kook Kim. "Distributed Training of Deep Learning Models Using Micro Batch Streaming." *KIPS Transactions on Software and Data Engineering*, 2023. doi:10.3745/KTSDE.2023.12.3.101.

4 **DoangJoo Synn** and Jong-Kook Kim. "Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders." *Proceedings of ACK 2021*, 2021. doi:10.3745/PKIPS.y2021m11a.446.

5 Byung Hoon Ahn, **DoangJoo Synn**, Masih Derkani, Eiman Ebrahimi, and Hadi Esmaeilzadeh. "Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection." *Neural Information Processing Systems Demonstrations*, 2021.